

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #12 - Blood Bank Inventory & Emergency Donor Matcher

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1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall cross-reference an emergency blood unit request against the real-time blood bank inventory and broadcast SMS alerts to compatible registered donors located within a 10 km radius of the requesting facility.	High	Pass: All compatible donors within 10 km receive the SMS within 30 s of the emergency flag being raised. Fail: Any donor with an incompatible blood group is alerted, or a compatible in-radius donor is alerted after 30 s.	Core capability of the system; shortens the time between shortage detection and donor arrival.
FR-002	Functional	The system shall allow the Blood Bank Manager to register an incoming blood unit by recording a unique bag ID, blood group, component type, donation timestamp and computed expiry date.	High	Pass: On saving, the unit appears in the inventory within 2 s with a system-computed expiry matching the component shelf-life rule (e.g. platelets = donation date + 5 days). Fail: A duplicate bag ID is accepted, or a mandatory field is saved empty.	Inventory accuracy is the precondition for every matching, allocation and expiry decision.
FR-003	Functional	The system shall monitor the shelf life of every stored unit, flag units expiring within 72 hours as "Near Expiry", and automatically move units past their expiry date to the "Quarantined" state so that they cannot be allocated.	High	Pass: A unit expiring in 71 h is listed under Near Expiry at the next hourly scan; a unit 1 minute past expiry is shown as Quarantined and cannot be selected. Fail: An expired unit is offered as available stock.	Prevents transfusion of expired components and reduces wastage by surfacing units in time to be used.
FR-004	Functional	The system shall allow a Donor to register a profile and shall automatically determine eligibility from age (18-65), weight (≥ 50 kg) and last donation date (≥ 90 days for whole blood), marking the donor as Eligible or Ineligible.	Medium	Pass: A donor who donated 60 days ago is stored as Ineligible with reason "minimum interval not met" and is excluded from every alert broadcast. Fail: An ineligible donor appears in a broadcast recipient list.	Alerting ineligible donors wastes response time in an emergency and risks donor health.
FR-005	Functional	The system shall allow an Emergency Requester to raise a request specifying blood group, component type, number of units and facility location, and shall return the matching available stock count together with a reservation ID for each unit held.	High	Pass: For a request of 2 units of O-negative PRBC with 3 units in stock, the response returns 2 reserved unit IDs and the available count drops to 1 within 3 s. Fail: More units are reserved than exist, or no reservation ID is returned on a match.	Gives the requesting clinician an immediate, actionable answer and locks the stock against competing requests.
NFR-001	Nonfunctional - Performance & Security	The inventory ledger shall maintain strict transactional consistency, preventing the same blood bag from being simultaneously allocated to two different requests.	High	Pass: Under a simulated peak load of 200 concurrent allocation requests on the same bag ID, exactly one succeeds and all others receive "unit already allocated", with median latency ≤ 500 ms. Fail: Any double allocation, or latency above 500 ms at peak.	A double-allocated unit means a patient is promised blood that does not exist - a safety-critical failure.
NFR-002	Nonfunctional - Security & Privacy	The system shall protect donor personal data by encrypting it at rest (AES-256) and in transit (TLS 1.3), enforcing role-based access control, and writing an immutable audit entry for every read of donor contact details.	High	Pass: A penetration test confirms no donor phone number is retrievable over plaintext HTTP; a Blood Bank Manager account cannot export the donor registry; every contact-detail access appears in the audit log with user ID and timestamp. Fail: Any unauthenticated or unlogged access to donor PII.	Donor contact and medical data are sensitive personal data; a leak would breach privacy law and destroy donor trust in the registry.

2. UML Use-Case Diagram

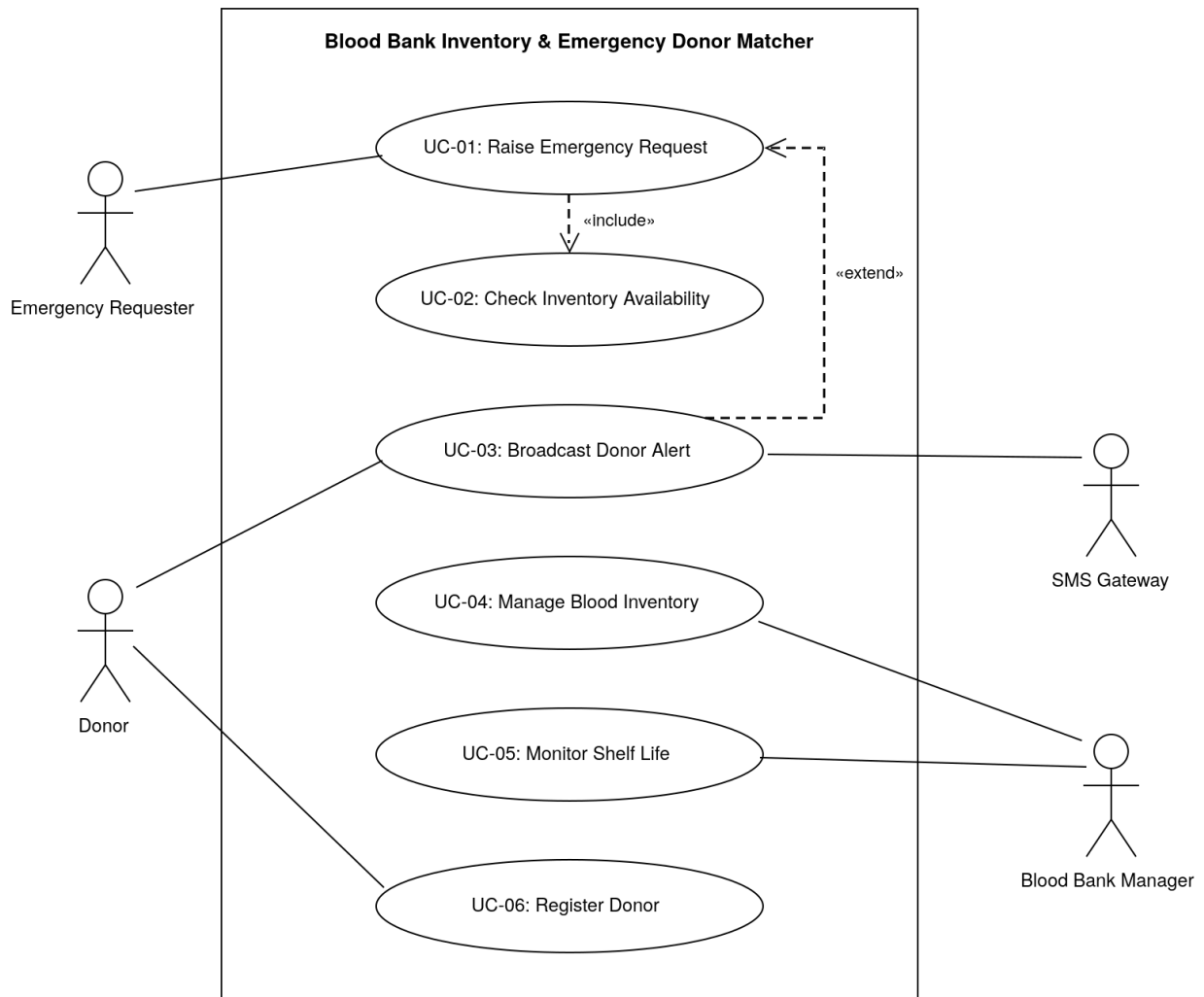


Figure 1: Use-case diagram - Blood Bank Inventory & Emergency Donor Matcher

«include» UC-01 includes UC-02 - an emergency request always checks live inventory before responding.

«extend» UC-03 extends UC-01 - the donor broadcast fires only when available stock is insufficient.

3. Use-Case Flow: UC-01 Raise Emergency Blood Request

Primary Actor	Emergency Requester (hospital / clinician)
Secondary Actors	Blood Bank Manager, Donor, SMS Gateway, Geolocation Service
Trigger	Requester submits an emergency blood requirement for a patient.

Preconditions

1. The Emergency Requester is registered and authenticated with a valid facility ID.
2. The blood bank inventory ledger is online and synchronised (last sync less than 60 s old).
3. At least one blood bank is mapped to the requester's geographic zone.
4. The donor registry contains donors with verified contact numbers and eligibility status.

Postconditions

Success guarantee

1. The requested units are reserved with a unique reservation ID and are no longer allocatable to any other request.
2. The available stock count is decremented atomically in the ledger.
3. The transaction is written to the immutable audit log with requester ID, timestamp and unit IDs.

Minimal guarantee (on any exit)

4. No unit is left in a partially allocated state; the request is recorded with a final status of Fulfilled, Partially Fulfilled, Escalated or Cancelled.

Main Success Scenario

1. The Emergency Requester selects "Raise Emergency Request" and the system displays the request form.
2. The Requester enters the blood group, component type (Whole Blood / PRBC / Platelets / Plasma), number of units, urgency level and facility location.
3. The Requester submits the request.
4. The system validates the input (valid blood group, units ≥ 1 , location resolvable).
5. The system raises an emergency flag and timestamps the request.
6. «include» UC-02 Check Inventory Availability - the system queries the real-time ledger for compatible, non-expired, non-quarantined units, applying blood-group compatibility rules.
7. The system finds sufficient matching units available.
8. The system locks the required units in a single atomic transaction (NFR-001) and generates a reservation ID.
9. The system displays the reservation ID, the reserved unit IDs, the holding blood bank and the collection window to the Requester.
10. The system notifies the Blood Bank Manager of the pending collection and writes the audit entry.
11. The use case ends successfully.

Alternate Flow 7a: Insufficient or No Matching Stock («extend» UC-03 Broadcast Donor Alert)

- 7a1. The system reserves whatever compatible units are available and marks the request Partially Fulfilled.
- 7a2. The system computes the shortfall (requested units minus reserved units).
- 7a3. The system queries the donor registry for donors who are Eligible, of a compatible blood group, and within a 10 km radius of the requesting facility (via the Geolocation Service).
- 7a4. The system dispatches SMS alerts to that donor list through the SMS Gateway within 30 seconds of the emergency flag (FR-001).
- 7a5. The system displays the shortfall and the number of donors alerted, and escalates the request to the Blood Bank Manager.
- 7a6. As donors confirm, the system updates the request status and notifies the Requester.
- 7a7. The flow resumes at step 10 for the units already reserved.

Alternate Flow 8a: Concurrent Allocation Conflict

- 8a1. Another request reserves one of the candidate units before this transaction acquires its lock.
- 8a2. The system rolls back the partial transaction so that no unit is double-allocated (NFR-001).
- 8a3. The system re-runs UC-02 once against the refreshed ledger.
- 8a4. If units are now available the flow resumes at step 8; otherwise it continues at 7a.